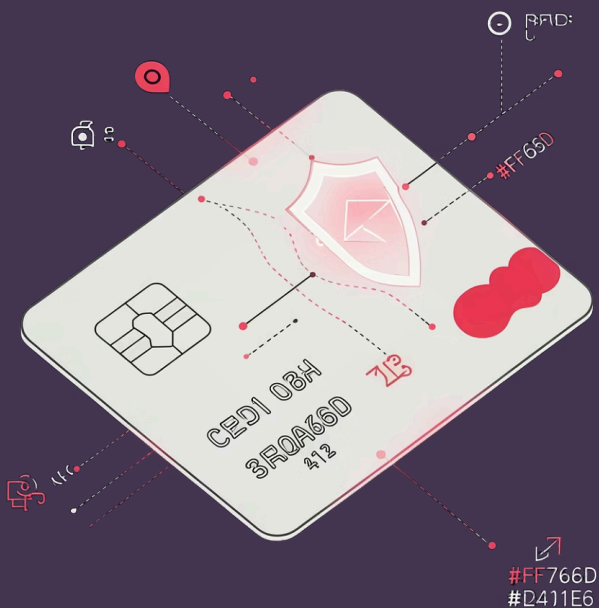




Credit Card Fraud Analysis Report

A data-driven intelligence briefing for fraud analytics teams — 2024–2025 threat landscape, ML adoption metrics, and actionable defense strategies.

The Fraud Crisis: €4.2B in Losses (2024)

€4.2B

European Payment Fraud

Total fraud losses in 2024, up from €3.5B in 2023 — a 20% year-over-year surge.

\$362B

Global Forecast (2023–2028)

Projected cumulative global online payment fraud losses over a five-year horizon.

\$1.1B

Enumeration Attack Impact

Follow-on fraud generated by enumeration attacks alone in a single year.



Emerging Fraud Patterns & Tactics



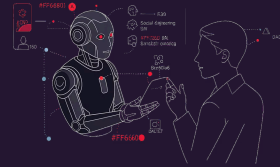
OTP Interception

Bypasses strong authentication, weaponized especially for digital wallet provisioning fraud.



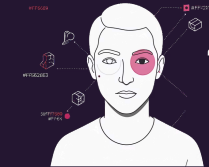
Provisioning Fraud & Malicious Apps

Threat actors exploit mobile vulnerabilities and POS system misconfigurations at scale.



AI-Enabled Social Engineering

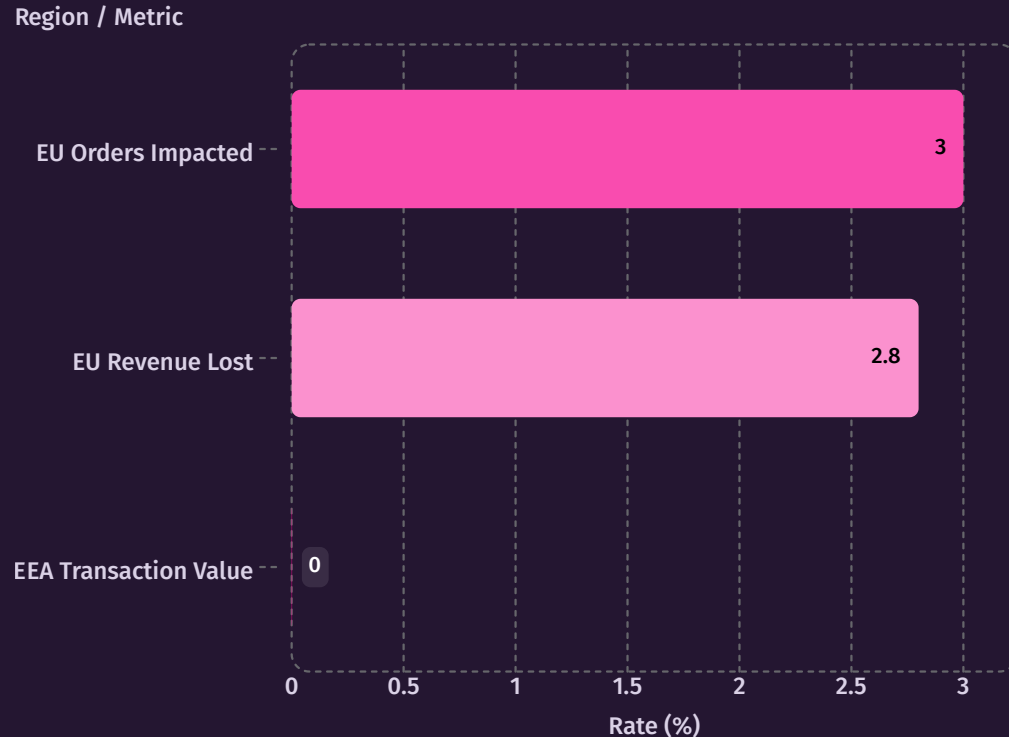
Scammers combine traditional manipulation tactics with AI tools to maximize victim impact.



First-Party Fraud Rising

Customers increasingly engaging in fraud, blurring the line between legitimate users and fraudsters.

Risk Assessment: Fraud Rates by Region



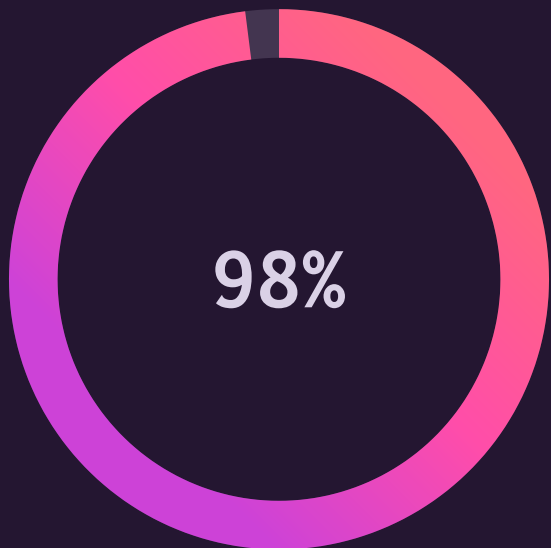
Key Regional Findings

Approximately **3% of European orders** are impacted by fraud, with **~2.8% of revenue** lost to fraudulent transactions annually.

The EEA payment fraud rate remains stable at **~0.002% of total transaction value** — a sign that Strong Customer Authentication (SCA) is holding the line on volume, even as fraud value climbs.

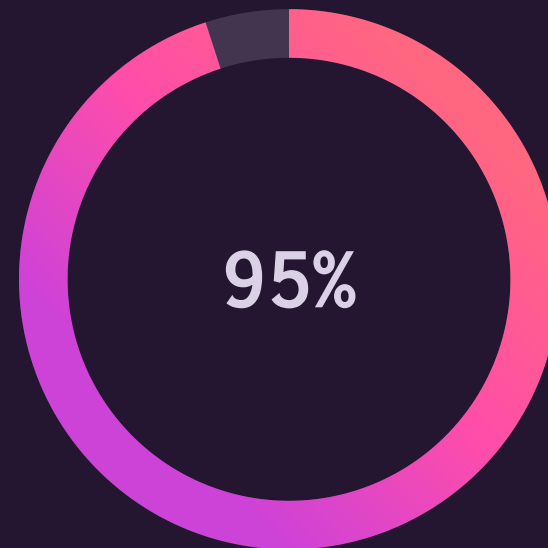
⚠ New manipulation tactics are circumventing SCA controls. Stability in rate does not mean stability in risk.

Machine Learning: The Detection Game-Changer



ML Integration

Of fraud and AML leaders already integrate ML into their daily workflows.



Detection Confidence

Express confidence in ML's ability to detect and prevent fraud effectively.

- **Real-Time Pattern Recognition:** ML identifies suspicious attributes invisible to human analysts
- **Supervised + Unsupervised:** Catch both known fraud patterns and emerging threats simultaneously

AI-Powered Fraud Prevention Strategies



Neural Networks & Random Forests

Ensemble methods improve detection accuracy across transaction types, reducing both false positives and missed fraud.



Anomaly Detection Systems

Flag unusual behavioral patterns before fraud occurs — catching threats at the behavioral layer, not just the transaction layer.



Continuous Model Retraining

Algorithms adapt to new fraud tactics in real-time, closing the window that fraudsters exploit after model deployment.



Data Collaboration

Organizations sharing risk intelligence to disrupt organized fraud networks before they can re-target across institutions.

The Dark Web Threat: Stolen Data Ecosystem

Volume vs. Quality Shift

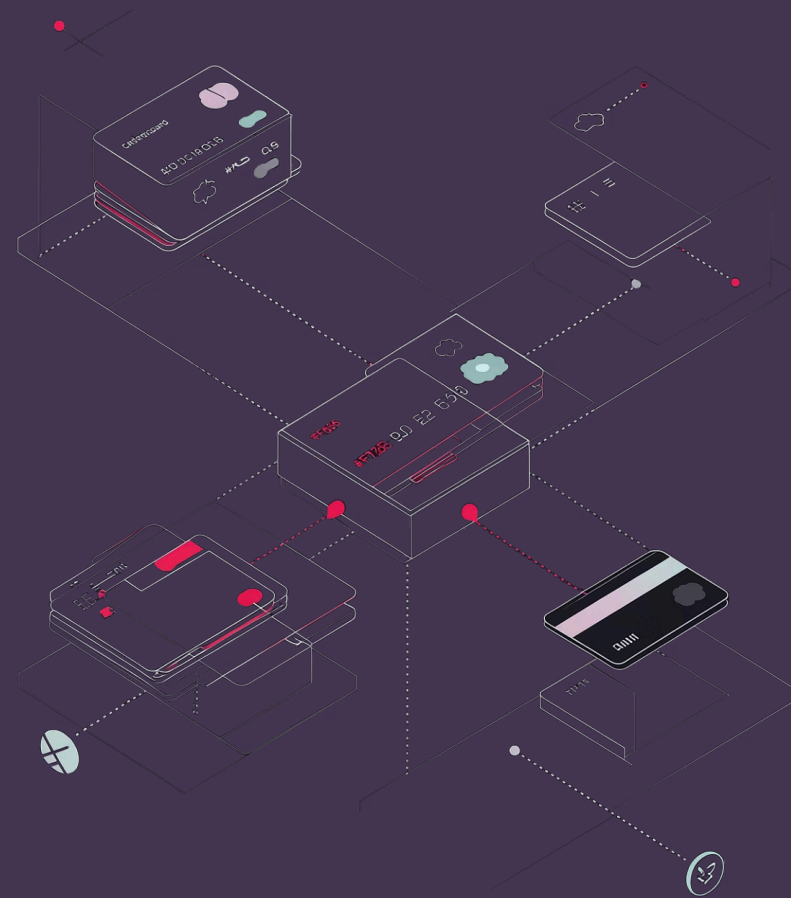
Raw stolen card data volume on dark web forums has **declined** — but attackers have compensated by targeting **higher-value cardholder attributes** like CVVs, billing addresses, and behavioral data.

- Industrial-scale e-skimmer infections and payment card testing services
- Mule networks facilitating global money laundering and fraud distribution

The Synthetic Identity Wave

AI-generated synthetic IDs and deepfakes are creating entirely new identity fraud vectors that traditional document verification systems cannot detect.

- Deepfake video bypassing liveness detection in KYC flows
- Synthetic identities combining real and fabricated data to pass bureau checks



Actionable Recommendations



Multi-Layer Defense

Combine ML detection, Strong Customer Authentication, and behavioral analytics for comprehensive protection. No single control is sufficient against today's adaptive adversaries.



Invest in Team Expansion

94% of fraud leaders are planning headcount increases in 2026 to manage rising complexity. Budget for talent now or risk capability gaps tomorrow.



Unify Data Architecture

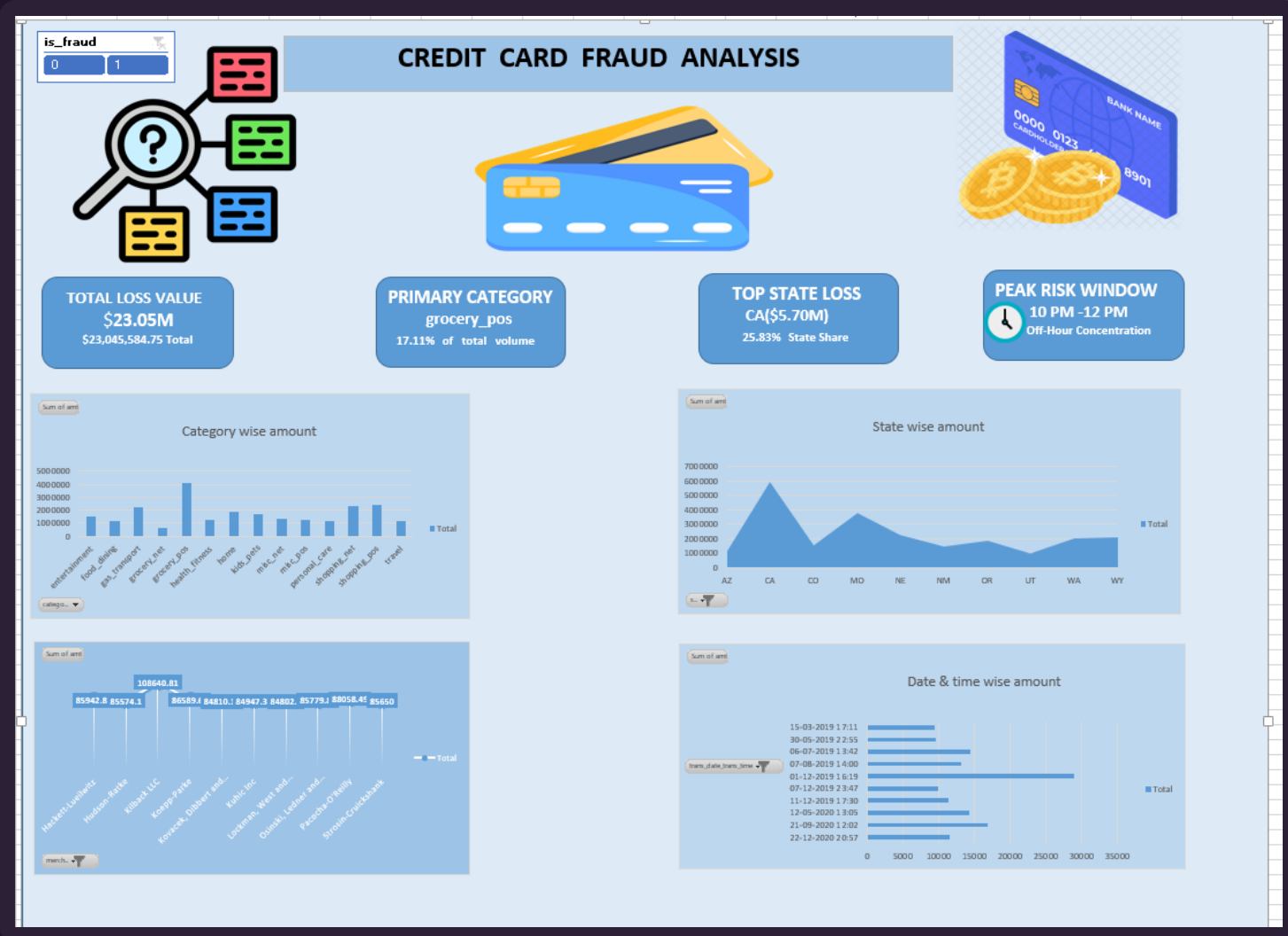
80% of fraud leaders struggle with a unified data view. Consolidating systems and data pipelines is the single highest-leverage infrastructure investment for fraud teams.



Monitor Emerging Vectors

Track OTP interception, provisioning fraud, and first-party fraud trends continuously. Threat intelligence feeds must be integrated directly into model retraining pipelines.

Dashboard: Transaction-Level Fraud Analysis



Dashboard Overview

This dashboard turns the transaction-level dataset into a compact fraud-risk view for business analysis.

KEY KPIS

- Total transaction amount
- Primary category
- Top state loss
- Peak risk window

CATEGORY VIEW

Compares transaction amount across merchant categories.

STATE VIEW

Highlights geographic differences in transaction amount.

TIME & MERCHANT

Shows concentration by date/time and merchant group.

Analytical dashboard derived from the transaction dataset used in this report.

Sources

- [Joint EBA-ECB report on payment fraud: strong authentication remains effective but fraudsters are adapting](#)
- [Ecommerce Fraud Rate Benchmarks \(2026\): Order Attack Rate, Losses & Risk Segments](#)
- [Machine Learning for Fraud Detection: How It Works — SEON](#)